

BOXUAN ZHANG

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📍 Rutgers University, New Brunswick, NJ - 08901

RESEARCH INTERESTS

My research focuses on *building trustworthy AI systems*, with an emphasis on **Agentic AI** and **LLM Reasoning**. I study how to make autonomous AI systems more reliable during deployment through runtime failure prediction, and how post-training can improve LLM reasoning while making model confidence better reflect correctness. I also work on reliable visual perception, including out-of-distribution detection and identifying AI-generated content.

- **Trustworthy Agentic AI:** I focus on making autonomous and multi-agent systems more reliable during deployment. My work focuses on auditing agent trajectories as they unfold so that failures are predicted and corrected before they propagate, and evaluating emerging risks such as uncontrolled self-replication.
- **LLM Post-Training and Reasoning:** I work on post-training methods that improve model reasoning, as well as on making a model's confidence better reflect the correctness of its reasoning.
- **Reliable Visual Perception:** I study how to determine when vision models can be trusted, with a focus on detecting out-of-distribution inputs and distinguishing AI-generated images from real ones.

EDUCATION

- **Rutgers University** 2025.09 - Present
Ph.D. in CS, Department of Computer Science Advisor: Prof. [Ruixiang \(Ryan\) Tang](#)
- **Wuhan University** 2022.09 - 2024.06
M.Eng. in AI, School of Computer Science Advisor: Prof. [Zengmao Wang](#) and Prof. [Bo Du](#)
- **Wuhan University** 2018.09 - 2022.06
B.Eng. in CS, School of Computer Science Advisor: Prof. [Jing Xiao](#)

INTERNSHIP EXPERIENCE

- **Advanced Micro Devices (AMD)** 2026.05 - Present
Research Scientist Intern, AMD GenAI Team Mentor: Dr. [Jiang Liu](#)
- **Shanghai Artificial Intelligence Laboratory** 2025.03 - 2025.07
Research Assistant, Center for Safe & Trustworthy AI Mentors: Dr. [Yi Yu](#) and Dr. [Dongrui Liu](#)

SELECTED PUBLICATIONS

* INDICATES EQUAL CONTRIBUTION

- **IJCV 2026 [Link]:** [Boxuan Zhang*](#), Jianing Zhu*, Zengmao Wang, Tongliang Liu, Bo Du, and Bo Han, "Unveiling Representation Expansion: A New Lens on Out-of-Distribution Detection". In *International Journal of Computer Vision* (IF = 10.3).
- **NeurIPS 2024 [Link]:** [Boxuan Zhang*](#), Jianing Zhu*, Zengmao Wang, Tongliang Liu, Bo Du, and Bo Han, "What If the Input is Expanded in OOD Detection?". In *Advances in Neural Information Processing Systems*.
- **ACL 2025 [Link]:** [Boxuan Zhang](#) and Ruqi Zhang, "CoT-UQ: Improving Response-wise Uncertainty Quantification in LLMs with Chain-of-Thought". In *Findings of the Association for Computational Linguistics*.
- **IEEE GRSL 2024 [Link]:** [Boxuan Zhang](#), Zengmao Wang, and Bo Du, "Boosting Semisupervised Object Detection in Remote-Sensing Images With Active Teaching". In *IEEE Geoscience and Remote Sensing Letters* (IF = 4.4).
- **FAGEN @ ICML 2026 [Link]:** [Boxuan Zhang*](#), Jianing Zhu*, Zeru Shi, Dongfang Liu, and Ruixiang Tang, "AgentForesight: Online Auditing for Early Failure Prediction in Multi-Agent Systems". In *Failure Modes in Agentic AI (FAGEN) Workshop @ International Conference on Machine Learning*.
- **arXiv 2026 [Link]:** [Boxuan Zhang](#), Jianing Zhu, Qifan Wang, Jiang Liu, and Ruixiang Tang, "Micro-Defects Expose Macro-Fakes: Detecting AI-Generated Images via Local Distributional Shifts".
- **arXiv 2025 [Link]:** [Boxuan Zhang*](#), Yi Yu*, Jiakuan Guo, and Jing Shao, "Dive into the Agent Matrix: A Realistic Evaluation of Self-Replication Risk in LLM Agents".

- **arXiv 2025 [Link]:** Zicong He*, **Boxuan Zhang***, Weihao Liu*, Ruixiang Tang, and Lu Cheng, “What Shapes a Creative Machine Mind? Comprehensively Benchmarking Creativity in Foundation Models”.
- **EMNLP 2026 [Link]:** Jiaqian Li, Yanshu Li, **Boxuan Zhang**, Ruixiang Tang, and Kuan-Hao Huang, “TRACES: Proactive Safety Auditing for Multi-Turn LLM Agents via Trajectory-State Modeling”. In *Conference on Empirical Methods in Natural Language Processing (Main)*.
- **ICME 2026 [Link]:** Yanchuan Tang*, Taowen Wang*, Yuefei Chen, **Boxuan Zhang**, Qiang Guan, and Ruixiang Tang, “Shifting Uncertainty to Critical Moments: Towards Reliable Uncertainty Quantification for VLA Model”. In *IEEE International Conference on Multimedia and Expo. (Best Paper Candidate)*
- **arXiv 2026 [Link]:** Minghao Guo*, Qingyue Jiao*, Zeru Shi*, Yihao Quan, **Boxuan Zhang**, Danrui Li, Liwei Che, Wujiang Xu, Shilong Liu, Zirui Liu, Mubbasir Kapadia, Vladimir Pavlovic, Jiang Liu, Mengdi Wang, Yiyu Shi, Dimitris N. Metaxas, and Ruixiang Tang, “MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory”.
- **Technical Report 2025 [Link]:** Shanghai AI Lab: Xiaoyang Chen, Yunhao Chen, ..., **Boxuan Zhang**, ... [30+ authors], “Frontier AI Risk Management Framework in Practice: A Risk Analysis Technical Report”.

SELECTED AWARDS AND HONORS

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| • 2025–2026 Excellent TA Award , Department of Computer Science, Rutgers University | 2026.06 |
| • Gold Reviewer , International Conference on Machine Learning (ICML 2026) | 2026.05 |
| • Best Paper Award Candidate , IEEE International Conference on Multimedia and Expo (ICME 2026) | 2026.03 |

SERVICES

Area Chair

AI-Native Academia Workshop @ Neural Information Processing Systems (NeurIPS)	2026
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Conference Reviewer

Neural Information Processing Systems (NeurIPS)	2025 - 2026
International Conference on Learning Representations (ICLR)	2026 - 2027
International Conference on Machine Learning (ICML)	2026
AAAI Conference on Artificial Intelligence (AAAI)	2027

Journal Reviewer

ISPRS Journal of Photogrammetry and Remote Sensing (IF = 12.2)	2024 - Present
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TEACHING

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| Teaching Assistant, Rutgers University | 2025.09 - Present |
| CS 344: Design and Analysis of Computer Algorithms | |
| CS 439: Introduction to Data Science | |
| Undergraduate Student Mentor, Wuhan University | 2022.09 - 2023.06 |
| Facilitated freshmen’s transition to university life at the School of Computer Science. | |

SKILLS

Programming: Python (PyTorch), C/C++, Linux, Git, $\LaTeX$

Languages: English (professional working proficiency), Chinese (native proficiency)